

Project Report

UNIFIED MILITARY ANALYTICS AND COMPARISON DASHBOARD



Submitted By: Deepak Avachitkar

Institute: LNCT Bhopal

Project Report Submitted To: Neha ma'am

Program Track: Data Analyst

Tool Used: Power BI

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Gmail: avachitkardeepak@gmail.com

Project link: springboardmentor9999x-rgb/Military-Analytics-Dashboard at [deepak-avachitkar](https://github.com/deepak-avachitkar)

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- Data Collection & Formatting Issues
- Visualizing Vastly Different Scales
- Complex Aggregation in the Coalition Builder

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Project Overview

Title: Unified Military Analytics and Comparison Dashboard

Problem Statement: Military and geopolitical data is often scattered, outdated, or locked on single platforms. This makes it difficult to compare countries, understand defence power distribution, or analyse multi-nation conflicts. Right now, analysts have to manually combine data, which is slow and inefficient. There is a need for a dynamic, interactive tool that brings together defence and economic data, allowing users to compare countries individually or test coalition strength against competitors.

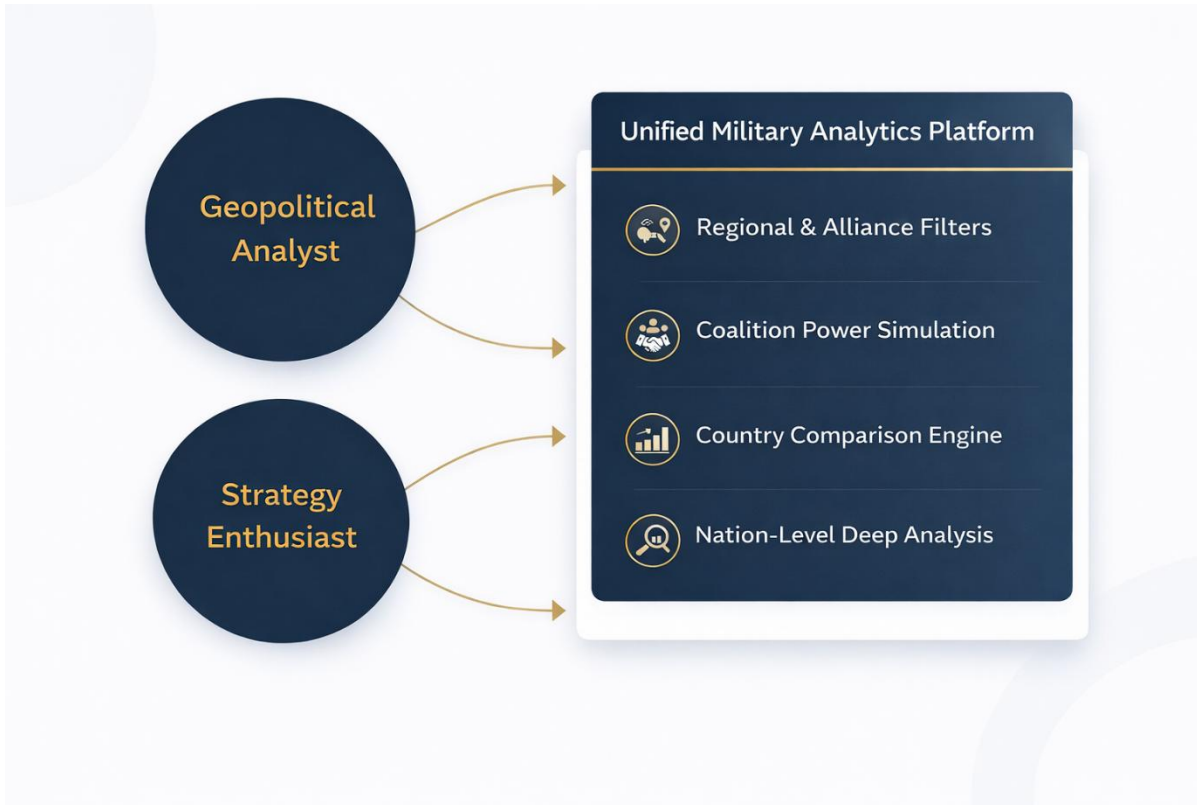
Objective: The goal is to build an interactive dashboard that shows military power across the world in 2025. The project will use Python pipelines to collect and clean open-source defence data from GlobalFirepower.com (covering 140+ countries). The processed data will then be visualized in Tableau, giving users a flexible platform to explore alliances, compare strengths, and track key performance indicators (KPIs).

Scope: The project covers the full cycle of data collection, processing, and visualization, including:

- **Web Scraping:** Automatically gather military data for 140+ countries using Python (requests, BeautifulSoup) from a prepared list of URLs.
- **Data Pipeline:** Clean and standardize the data, fix missing values, and organize 50+ defence and economic indicators.
- **KPI Engineering:** Create advanced metrics such as Power Index Rank Gap, Assets per Capita, and Defence Budget-to-GDP ratio.
- **Dashboard Development:** Build four connected modules:
 - **Quick Stats:** Fast insights into global military data.
 - **Nation Overview:** Detailed profile of each country.
 - **Compare Powers:** Side-by-side comparison of countries.

- Coalition Builder: Test combined strength of alliances against competitors.

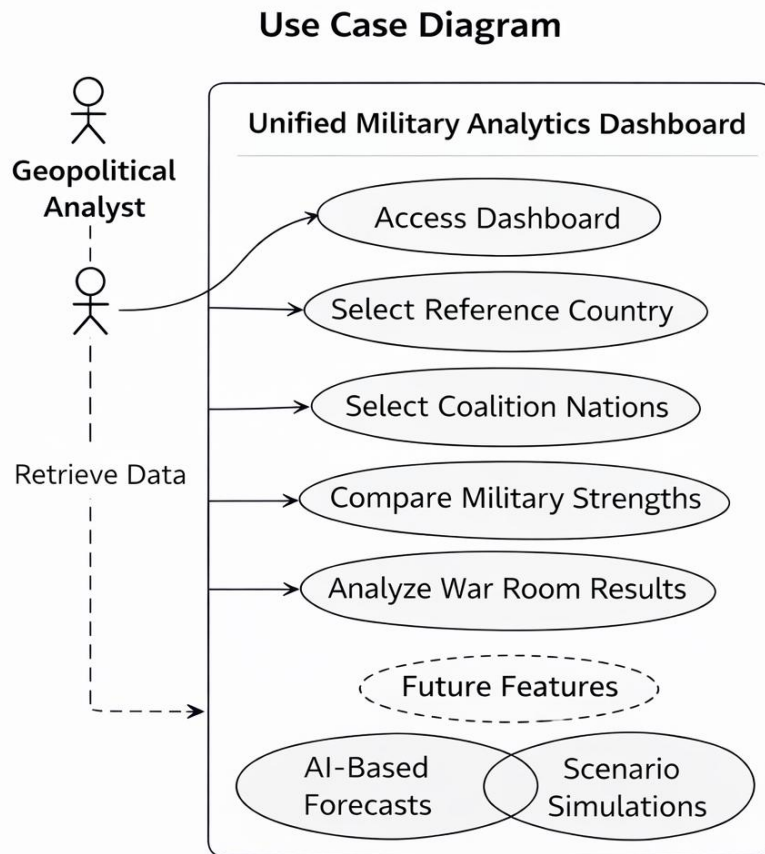
Target Users:



- Geopolitical Analysts: Need fast access to compare military and economic data across nations.
- Defence Enthusiasts: Explore alliance strength and simulate war-room scenarios.
- Analytics Recruiters: Review ETL pipelines, Tableau logic, and dashboard design skills.

Expected Outcomes

- A complete dashboard suite with filters, dropdowns, and navigation.
- Automated scraping system and cleaned dataset.
- GitHub repo with all code, data, and documentation.



2. Data Source & Collection

Data Source

The dataset was built by scraping public military metrics from GlobalFirepower.com, which tracks defence strength for 140+ nations. A control file (*links_for_military_data.txt*) was used to manage all target URLs for consistency.

Data Collection

A custom Python script (*scrape_military_metrics.py*) automated the process:

1. URL Ingestion: Reads each country's profile link from the control file.
2. Request & Validation: Sends HTTP requests and ensures a $\geq 95\%$ success rate.
3. Parsing: Extracts key metric blocks (Manpower, Airpower, Financials, etc.).
4. Storage: Saves raw data into *military_raw_data.csv* and temporary HTML files for debugging.

```
3]: # User-Agent header to avoid basic blocking
HEADERS = {"User-Agent": "Mozilla/5.0"}
```

Function: read_links_txt

Reads metric URLs from a TXT file

Cleans bullets/arrows and returns valid URLs

```
5]: def read_links_txt(path):
    links = []
    with open(path, "r", encoding="utf-8") as f:
        for line in f:
            line = line.strip()

            if not line:
                continue

            # Remove bullets and arrow symbols if present
            line = line.replace("-", "")
            if ">" in line:
                line = line.split(">")[0].strip()

            # Keep only valid URLs
            if line.startswith("http"):
                links.append(line)

    return links
```

Tools & Libraries Used

- Python 3.x – Core language for ETL pipeline.
- Requests – Fetches raw HTML via HTTP GET.
- BeautifulSoup (bs4) – Parses HTML to extract metrics.
- Pandas & NumPy – Structures data into DataFrames and handles initial cleaning.

Challenges

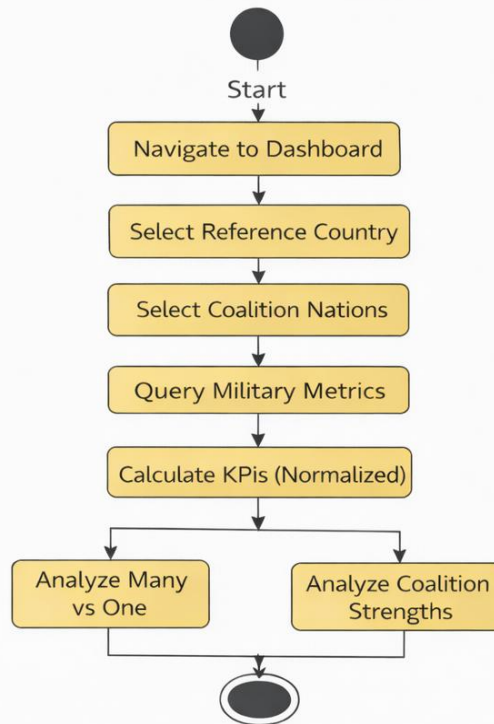
- Formatting Issues: Raw data contained symbols like \$, %, +, and commas, requiring extra string cleaning.
- Validation Needs: Ensuring all 140+ countries had complete metric blocks meant adding checks for missing or null values.

3. Data Pipeline Architecture

System Architecture

The project implements a linear four-stage ETL and visualization pipeline optimized for cross-platform deployment. All extraction, cleaning, and feature engineering steps are consolidated within a single Jupyter Notebook (scrape_military_metrics.ipynb) to streamline execution and reproducibility.

Activity Diagram



Pipeline Stages:

1. Scraping

- Raw HTML content is retrieved using Python requests.
- BeautifulSoup parses the DOM to isolate metric blocks.
- Initial output stored as `military_raw_data.csv`.

2. Cleaning

- Data normalization and null handling performed with Pandas.
- Intermediate datasets (`military_cleaned.csv`, `military_standardized.csv`) staged for validation and quality checks.

3. KPI Engineering

- Advanced calculated fields (e.g., *Power Index Rank Gap*, *Assets per Capita*) generated in final notebook cells.
- Golden dataset exported as `military_final.xlsx`.

4. Visualization

- Final dataset ingested by Tableau.

- Interactive dashboards built with unified filters, dropdowns, and navigation controls.

4.Data Cleaning & Preprocessing

Raw Dataset:

The initial scrape produced `military_raw_data.csv`, a flat file with string-heavy values. Since Tableau requires strict typing for accurate aggregation, a Pandas-based cleaning pipeline was implemented.

Cleaning Workflow

1. Symbol Removal – Currency signs, commas, plus signs, and percentage markers stripped from raw values.
2. Type Conversion – Asset/manpower metrics cast to int; macroeconomic fields (GDP, defense budgets) cast to float.
3. Missing Values – Nulls handled via imputation or removal, ensuring <2% missing data threshold.
4. Column Standardization – Headers normalized to snake_case (e.g., `total_aircraft`, `defense_budget_usd`).
5. Validation – Intermediate files (`military_cleaned.csv`, `military_standardized.csv`) staged for QA checks to confirm structural integrity.

```
def clean_numeric(series):  
    return (  
        series.astype(str)  
        .str.replace(r"[,%+]", "", regex=True)  
        .str.replace(" ", "")  
        .replace("nan", np.nan)  
    )  
  
for col in df.columns:  
    if df[col].dtype == "object":  
        df[col] = clean_numeric(df[col])
```

Final Output:

The cleaned dataset was consolidated into `military_final.xlsx`, serving as the golden source for Tableau ingestion and KPI engineering.


```

40 total_merchant_marine_fleet      145 non-null    int64
41 railway_coverage_km             145 non-null    int64
42 roadway_coverage_km             145 non-null    int64
43 oil_production_bbl              145 non-null    int64
44 oil_consumption_bbl             145 non-null    int64
45 proven_oil_reserves_bbl         145 non-null    float64
46 natural_gas_production_cum      145 non-null    float64
47 natural_gas_consumption_cum     145 non-null    float64
48 proven_natural_gas_reserves_cum 145 non-null    float64
49 coal_production_cum             145 non-null    int64
50 coal_consumption_mt             145 non-null    int64
51 proven_coal_reserves_cum        145 non-null    float64
52 total_land_area_sq_km          145 non-null    int64
53 coastline_coverage_km          145 non-null    int64
54 border_coverage_km             145 non-null    int64
55 waterway_coverage_km           145 non-null    int64
dtypes: float64(10), int64(45), object(1)
memory usage: 63.6+ KB

```

```
[37]: df.to_csv("military_1cleaned.csv", index=False)
```

5.Dataset Structure

1. Primary Key Structure

The dataset is a cross-sectional snapshot of global military capabilities for 2025.

Primary Key: country (Text)

Each country appears once, ensuring unique national profiles.

Benefits:

- Prevents duplication
- Enables accurate slicer filtering
- Ensures correct DAX aggregations
- Supports stable table relationships

Future extension: If multi-year data is added, use Country + Year as a composite key.

2. Region & Alliance Mapping

To enable macro-level geopolitical analysis, categorical dimensions were added.

Region Structure:

- Continent
- Region

Used for:

- Drill-down analysis
- Regional comparisons
- Map visualizations

Hierarchy:

Continent → Region → Country

Alliance Structure:

- NATO_Status (NATO / Non-NATO / Neutral)
- Alliance_Block (NATO, BRICS, SCO, Independent)

Used for:

- Coalition filtering
- Bloc comparisons
- Dynamic KPI calculations

3. Core Dataset Schema (Key Indicator Groups)

Identification:

Country, Continent, Region, Alliance categories

Economic:

Defence Budget, GDP, Military % of GDP

Manpower:

Active, Reserve, Paramilitary forces

Military Assets:

- Land: Tanks, armored units, artillery
- Air: Aircraft, fighters, helicopters
- Naval: Carriers, submarines, surface fleet

Infrastructure:

Airports, ports, roads, railways

4. Power BI Model Design

Recommended structure: Star Schema

Dim_Country → Fact_Military_Stats (1-to-many)

(Currently implemented as a single flat analytical table)

5. Analytical Capabilities Enabled

Country vs Country benchmarking

Alliance vs Alliance comparison

Regional dominance analysis

Military density metrics
KPI-driven dashboards
Drill-through country profiles

6. Design Strengths

- Clean primary key
- No aggregation errors
- BI-optimized dimensions
- High performance in Power BI

6.KPI Engineering

Purpose of KPIs

Raw military totals do not allow fair country comparison due to differences in population, economy, and geography.

Therefore, KPIs were engineered as normalized ratio metrics to convert absolute values into comparable performance indicators for strategic analysis.

Static vs Dynamic KPI Design

Static KPIs (Pre-calculated Columns)

Computed during data preparation for fast performance:

- Assets per Capita
- Defence Budget to GDP Ratio
- Personnel Density
- Armor Intensity Index
- Military Burden Index

Dynamic KPIs (DAX Measures in Power BI)

Calculated in real time based on slicers:

- Coalition Strength Index
- Power Rank Gap
- Bloc-level totals

Implemented KPIs (Formula + Meaning)

KPI	Formula	Purpose
Assets per Capita	$(\text{Aircraft} + \text{Tanks} + \text{Naval Assets}) / \text{Population}$	Militarization density
Budget to GDP	Defence Budget / GDP	Economic military burden
Personnel Density	Total Personnel / Population	Mobilization level
Budget per Soldier	Defence Budget / Active Personnel	Investment per troop
Power Rank Gap	Selected Rank – Reference Rank	Relative strength
Air Power Ratio	Aircraft / Active Personnel	Air dominance focus
Armor Intensity	Tanks / Land Area	Ground defence density
Naval Strength	Naval Assets / Coastline	Maritime protection
Military Burden	Defence Budget / Population	Per-citizen cost
Coalition Strength (Dynamic)	Sum of total assets	Alliance power simulation

7. Dashboard Design & Architecture

The Unified Military Analytics platform is structured into four focused dashboard modules, each progressing from high-level global insights to detailed strategic comparison and coalition simulation.

This modular approach avoids visual clutter and supports step-by-step analytical decision making.

1. Quick Stats Dashboard

Objective: Provide a global military snapshot at a glance.

Key Features:

- Top country rankings
- Global totals (GDP, Assets, Defence Budget, Military Burden)
- Top manpower and asset distributions

KPIs Used:

- Total Global Assets
- Total Defence Budget
- Total GDP
- Military Burden Index
- Active Personnel

Purpose: Fast situational awareness and global power balance.

2. Nation Overview Dashboard

Objective: Deep-dive analysis of a single country's military profile.

Key Features:

- Country-specific KPI cards (rank, GDP, population, purchasing power)
- Personnel breakdown (active & reserve)
- Asset composition (air, land, naval)
- Infrastructure & coastline coverage visuals

KPIs Used:

- Active Personnel
- Asset Distribution
- Power Rank
- Defence & Economic Indicators

Purpose: Micro-level national capability assessment.

3. Compare Powers Dashboard

Objective: Direct side-by-side military benchmarking between two countries.

Key Features:

- Dual country selectors
- Comparative KPI cards
- Budget per soldier
- Total assets & manpower comparison

KPIs Used:

- Defence Budget
- Budget per Soldier
- Total Assets
- Manpower
- GDP

Purpose: Tactical 1-vs-1 strength evaluation.

4. Coalition Builder Dashboard

Objective: Simulate combined military strength of multiple countries against a reference power.

Key Features:

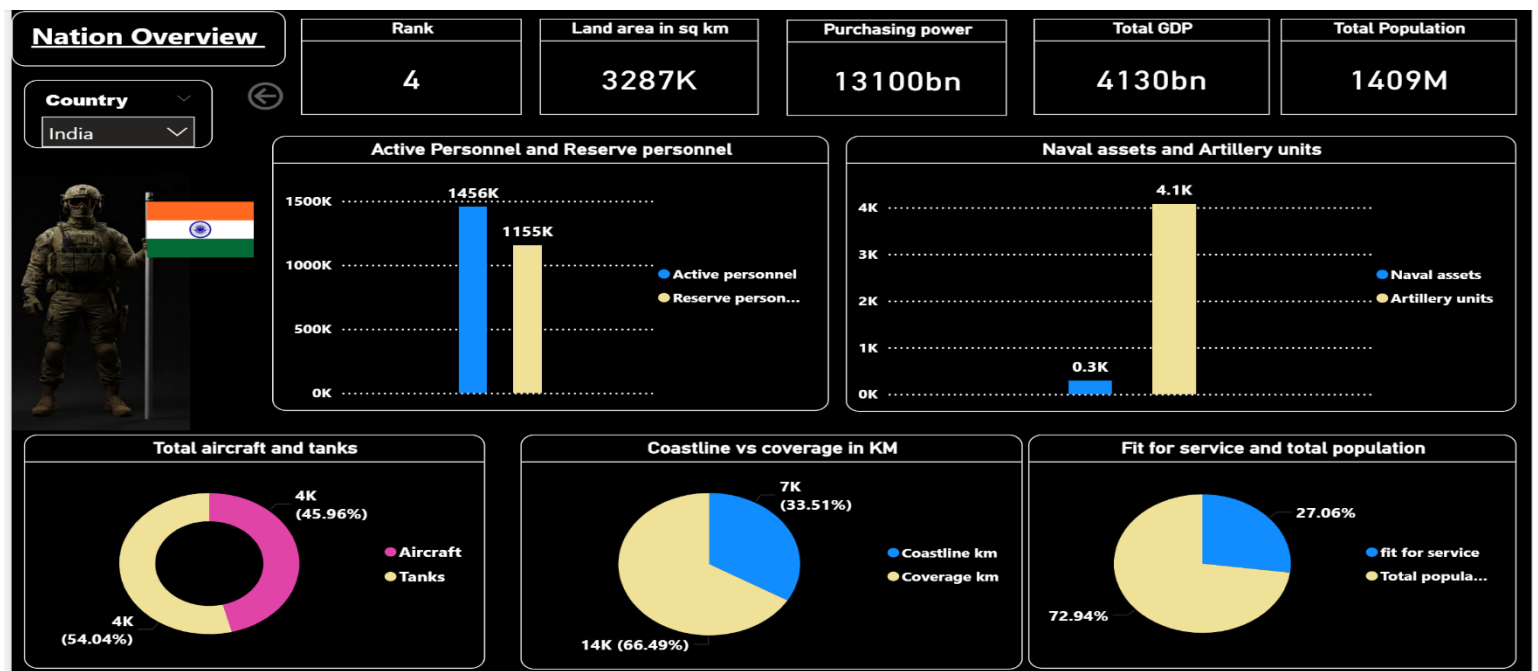
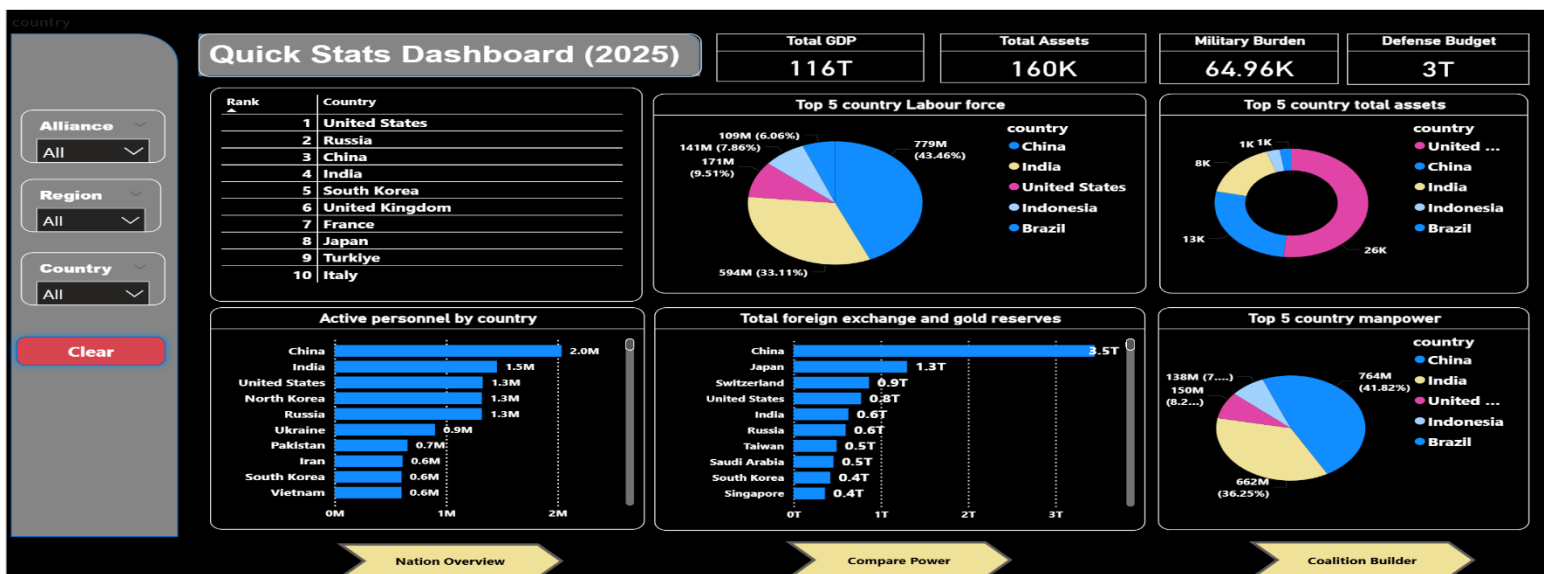
- Multi-country coalition selection
- Asset aggregation

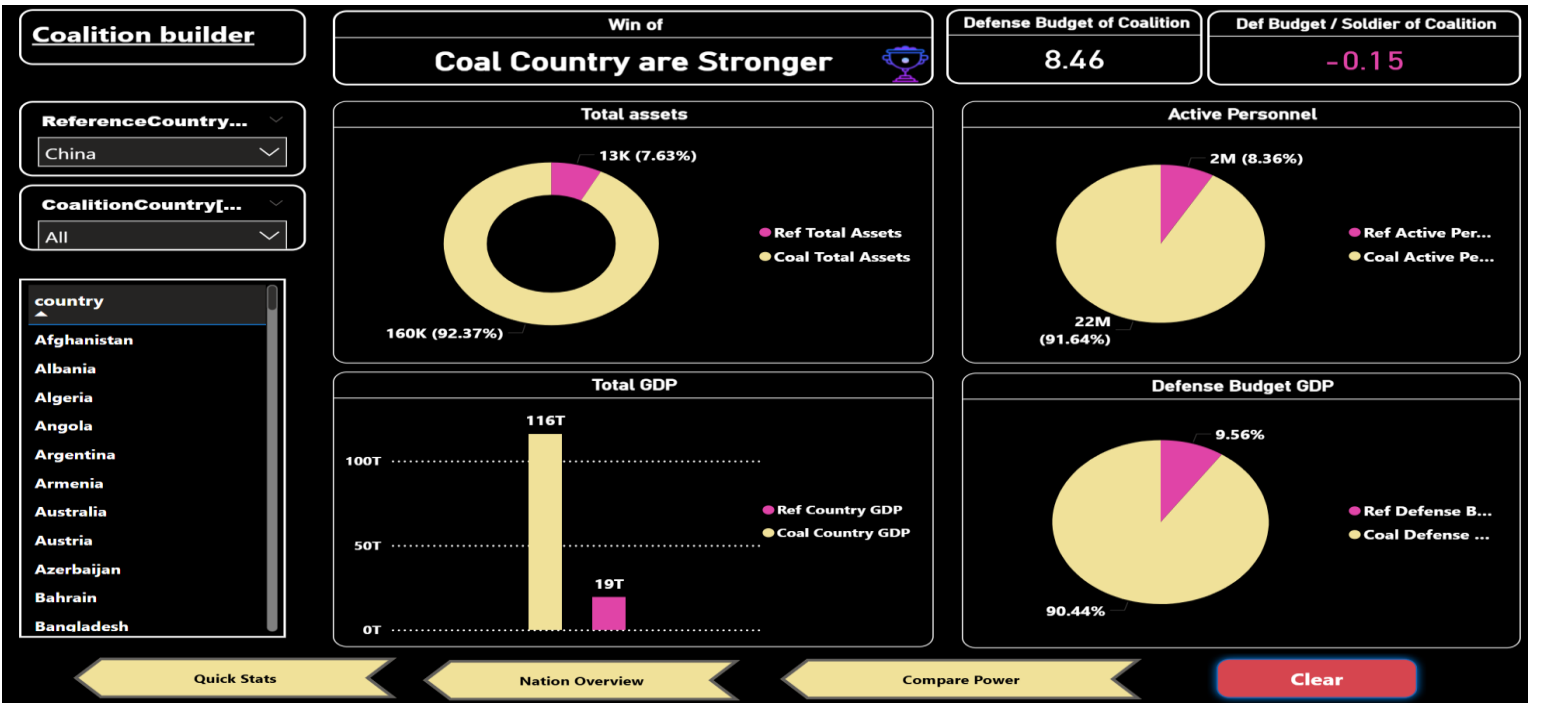
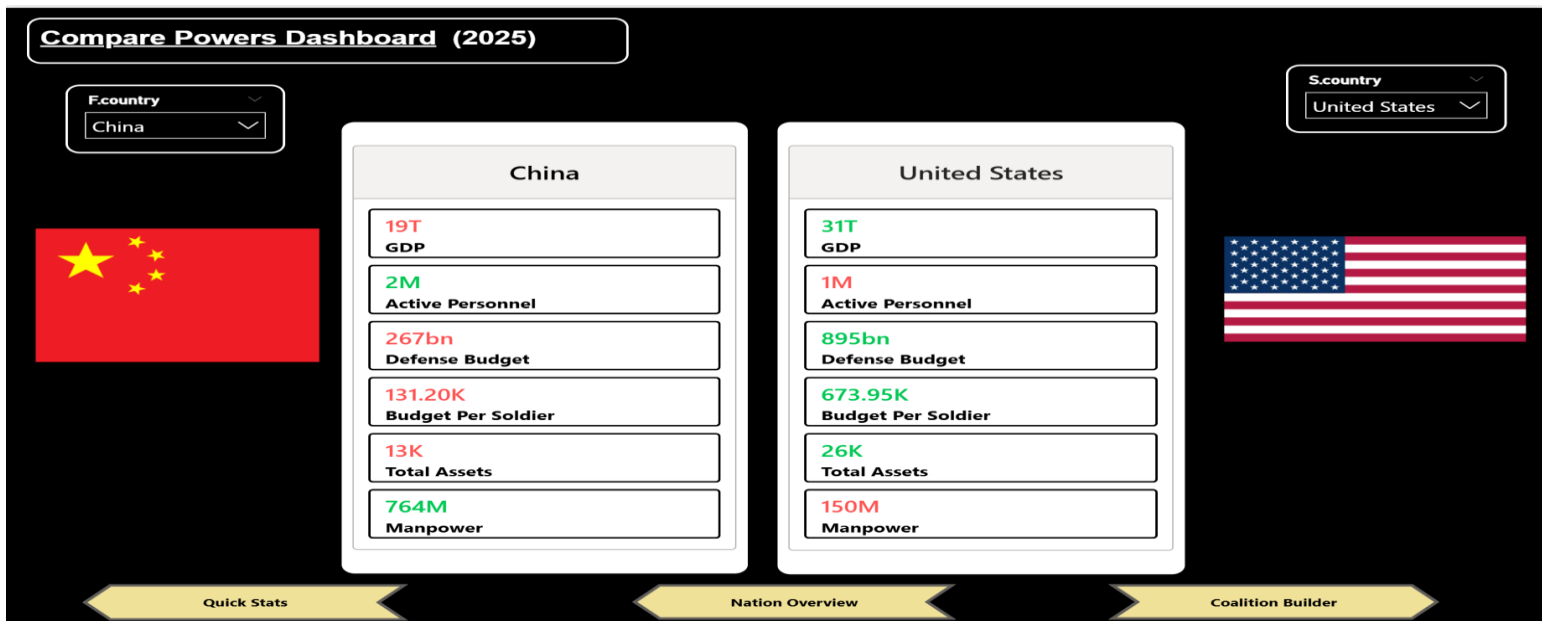
- Budget and manpower comparison
- Win/strength indicator

KPIs Used:

- Coalition Strength Index (total assets)
- Combined Defence Budget
- Coalition vs Reference manpower gap
- Budget efficiency comparison

Purpose: Strategic alliance scenario modelling





Filter & Interaction Design

A unified filtering system was implemented to maintain high interactivity with optimal performance.

- **Global slicers:** Region, Continent, and Alliance act as synchronized filters across all dashboards.
- **Analytical selectors:** Dedicated country selectors drive comparisons and coalition simulations dynamically.

Navigation Flow: Standard report tabs were replaced with custom navigation buttons to create an app-like experience while preserving filter context across pages.

8. Tool Implementation (Power BI)

Data Connection & Architecture

The finalized dataset (military_final.xlsx) was imported into **Power BI** as the core data model.

- Data types were validated (text for dimensions, numeric for measures).
- Geographic fields (Country, Region, Continent) were properly categorized for map visuals.
- Numeric measures were formatted as whole numbers or decimals to ensure accurate aggregations.

The model was optimized for performance using a clean, single-source analytical structure.

Interactive Controls

Dedicated selection mechanisms were implemented to enhance interactivity beyond basic slicers:

- **Country Selector (Nation Overview):** Enables isolated country profiling.
- **Dual Selectors (Compare Powers):** Two independent country selections for side-by-side benchmarking without filter conflict.
- **Reference Selector (Coalition Builder):** Allows selection of a baseline country to compare against a custom coalition.

DAX Measures & Logic

KPI formulas were implemented as calculated columns and DAX measures.

- **Selection Logic:** Measures dynamically respond to chosen countries using filter context.
- **Dynamic Aggregation:** Coalition Builder aggregates total assets and budgets across multi-selected countries.
- **Conditional Formatting:** Win/Loss indicators use DAX logic to dynamically change text and color based on strength comparison.

Dashboard Layout & UI Design

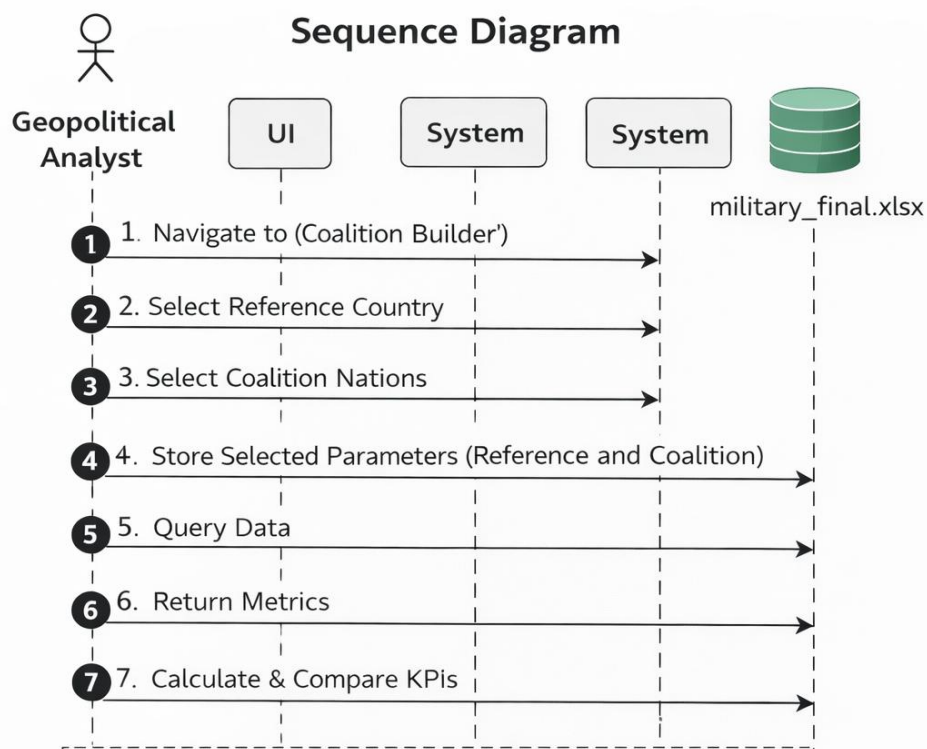
The dashboard layout was structured for clarity, responsiveness, and analytical focus.

- **Responsive grid layout:** Visuals were aligned using structured containers to maintain consistency across screen sizes.
- **Thematic consistency:** A dark-mode design with high-contrast accent colors was applied across all dashboards to enhance readability and create a strategic command-style interface.
- **Interactive navigation:** Custom navigation buttons and action-based interactions enable seamless movement between dashboards while preserving filter context.

9. Coalition Builder Logic

Purpose

The Coalition Builder works as a strategic simulation tool that allows users to combine multiple countries into a custom alliance and compare their total military strength against a selected superpower.



Selection Structure

Two interaction layers were used:

- **Reference Country Selector:** Single country chosen as the benchmark power
- **Coalition Selector:** Multi-country slicer used to form a custom alliance

This enables **many-vs-one comparison** without filter conflicts.

Aggregation Logic (Dynamic KPIs)

Power BI automatically aggregates selected coalition countries using DAX measures:

- **Coalition Total Assets** = Sum of aircraft + tanks + naval assets
- **Coalition Defence Budget** = Sum of defence budgets
- **Coalition Manpower** = Sum of personnel

All values update instantly based on coalition selection.

Comparison & Win/Loss Logic

DAX measures isolate the reference country and calculate:

- Asset gap (Coalition – Reference)
- Budget gap
- Manpower gap

A conditional indicator dynamically shows:

- **Coalition Stronger** or **Reference Stronger**
with automatic colour change for quick interpretation.

10. Testing & Validation

Before deployment, the dashboards were thoroughly tested to ensure accurate calculations, stable interactions, and smooth performance.

Testing Approach

- **KPI validation:** Key metrics were manually cross-checked against the raw Excel dataset to confirm calculation accuracy.
- **Edge case testing:** Countries with zero or extreme values were tested to ensure measures handled blanks and division scenarios correctly.
- **Performance testing:** Dashboard responsiveness was verified during heavy filtering and page navigation.

Test Category	Test Case Description	Status
Data Integrity	Null values handled properly	Passed
Data Integrity	All KPIs formatted correctly	Passed
Interactivity	Global slicers update all visuals	Passed
Interactivity	Compare selectors update KPIs instantly	Passed
Logic (Compare)	Rank gap works correctly	Passed
Logic (Coalition)	Coalition aggregation accurate	Passed
Logic (Coalition)	Win/Loss updates dynamically	Passed
UI/UX	Navigation keeps filters active	Passed
Performance	Dashboard loads fast	Passed

11. Challenges & Solutions

1. Data Cleaning & Formatting

Challenge:

Raw data contained mixed text and numbers (e.g., “\$3B”, “1.5 million”) and missing values that broke consistency.

Solution:

Numeric values were standardized using string cleaning and type conversion in Python/Pandas, with error handling to prevent pipeline failures.

2. Scale Imbalance in Visual Comparisons

Challenge:

Large countries dominated charts, making smaller nations visually insignificant in comparisons.

Solution:

Normalized KPIs (per capita and ratio-based metrics) were introduced to enable fair, readable comparisons across different scales.

3. Dynamic Coalition Aggregation Logic

Challenge:

Combining multiple countries dynamically and comparing them to a separate reference country required complex filter handling.

Solution:

Independent selectors and DAX measures were used to isolate the reference country while aggregating coalition totals accurately in real time.

12. Future Improvements

While the current dashboard provides a strong 2025 decision-support framework, the following enhancements can elevate it to an enterprise-level intelligence platform:

- **API-Based Data Integration:** Replace static datasets with automated API-driven refresh for near real-time updates.
- **Multi-Year Time Series Expansion:** Integrate historical data to enable trend analysis, growth tracking, and long-term power shift evaluation.
- **Predictive Analytics Integration:** Apply machine learning models to forecast defence budgets, asset growth, and military strength trends.

- **Web Deployment with Access Control:** Deploy as a secure web-based Power BI solution with role-based access control (RBAC) for restricted strategic simulations.
- **Advanced Scenario Modelling:** Enhance the Coalition Builder with additional variables such as logistics, geographic distance, and strategic capabilities.

13. Conclusion

Project Summary

The Unified Military Analytics Dashboard transformed unstructured web data into a structured decision-support system using an automated Python data pipeline and Power BI visual analytics. Military and economic indicators for 140+ countries were cleaned, standardized, and modelled for accurate comparison.

Advanced KPIs such as Military Burden Index and Assets per Capita revealed how economic capacity and geography shape military strength beyond raw asset counts. Dynamic dashboards enabled instant country comparisons and real-time coalition strength simulations, uncovering strategic advantages and vulnerabilities.

Overall, the project delivers a scalable, interactive intelligence platform that converts complex defence data into clear, actionable insights and provides a strong foundation for future predictive analytics and multi-year trend analysis.